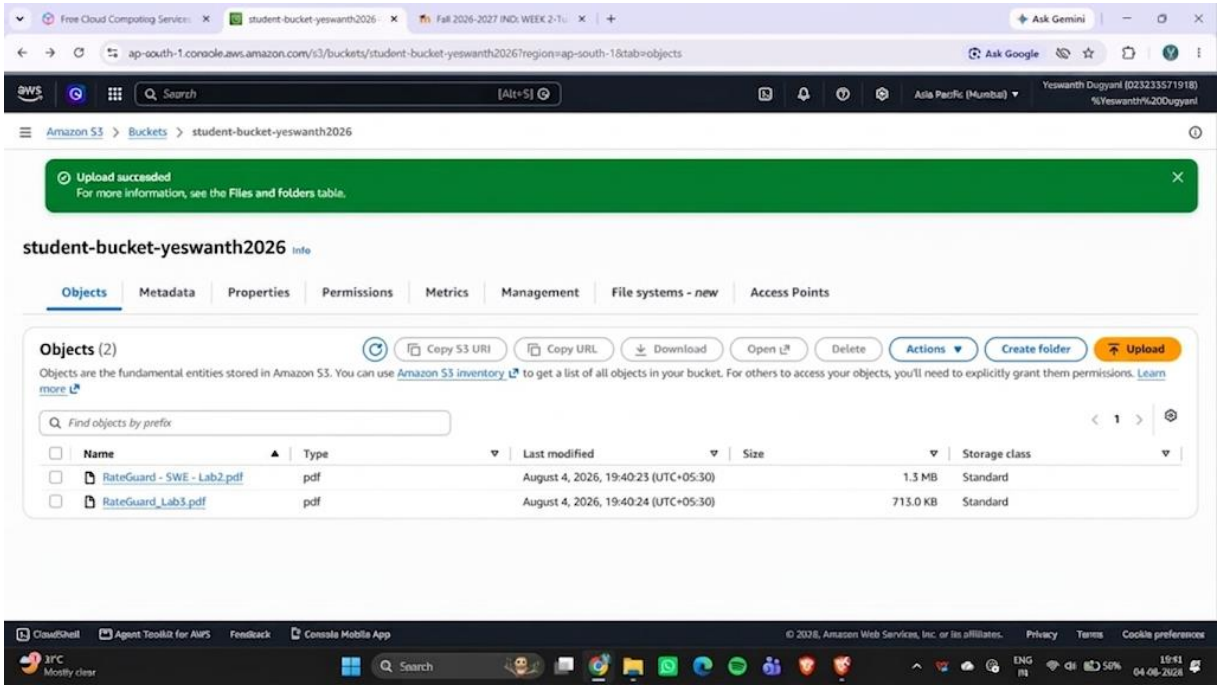
Cloud Architecture Design

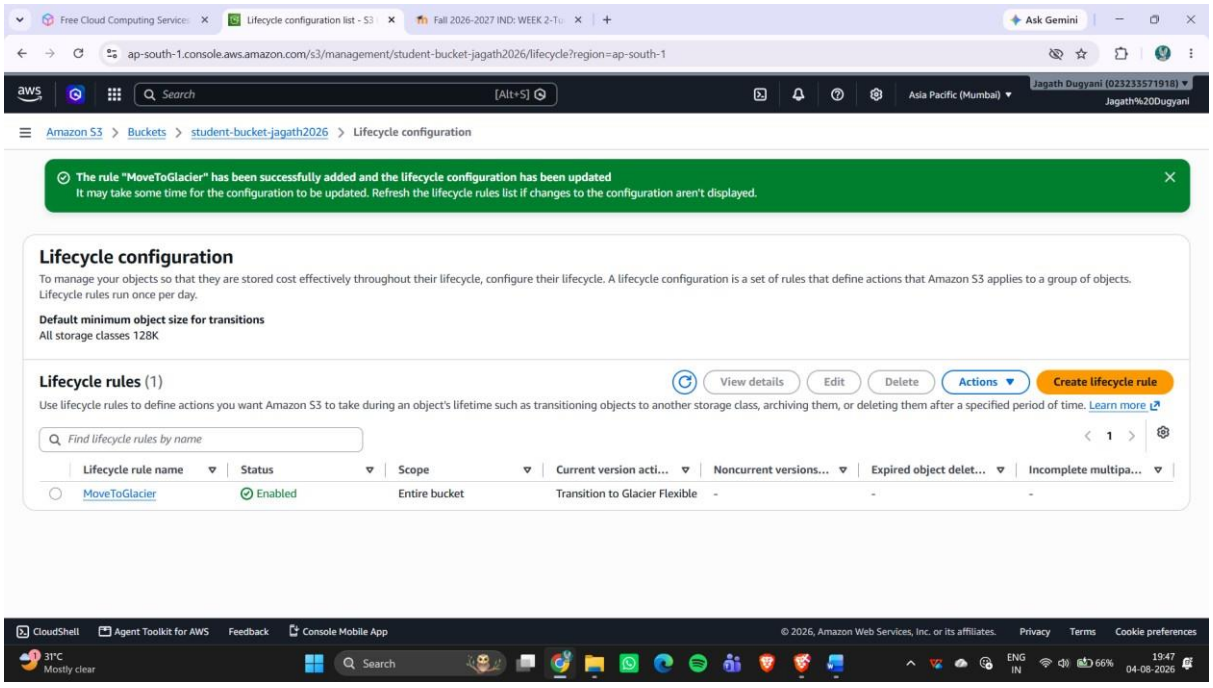
24BRS1170(Yeswanth)

DA – 1:

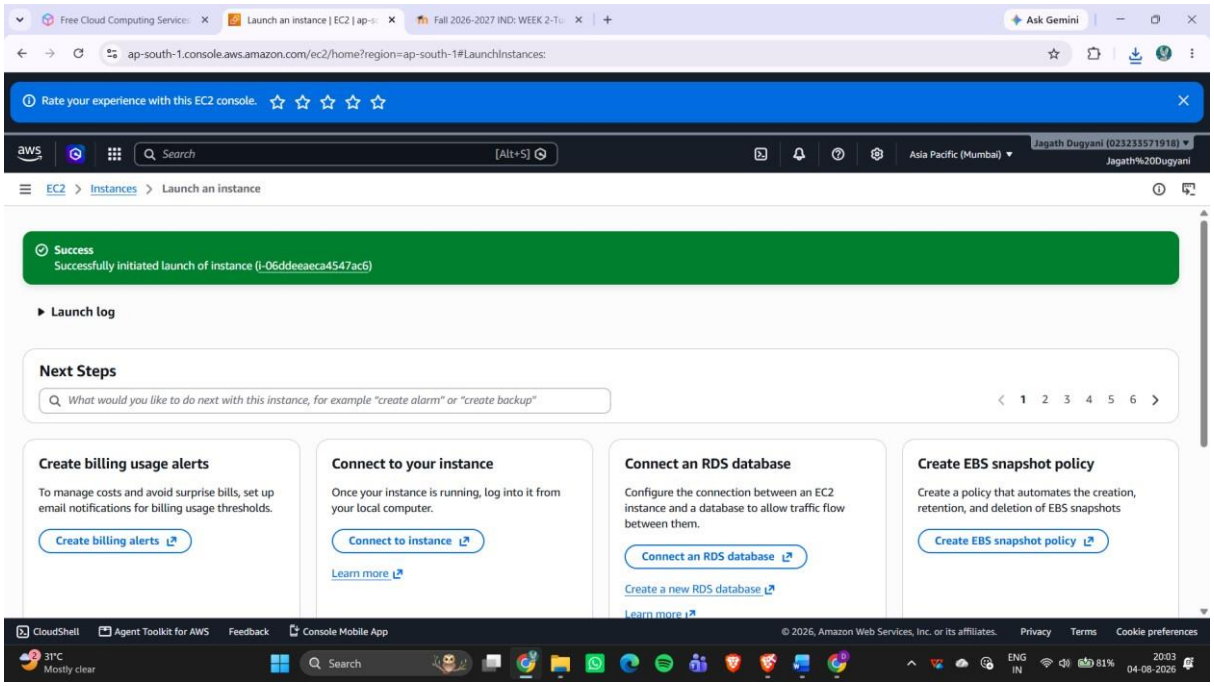
Amazon S3 Bucket Creation and File Upload



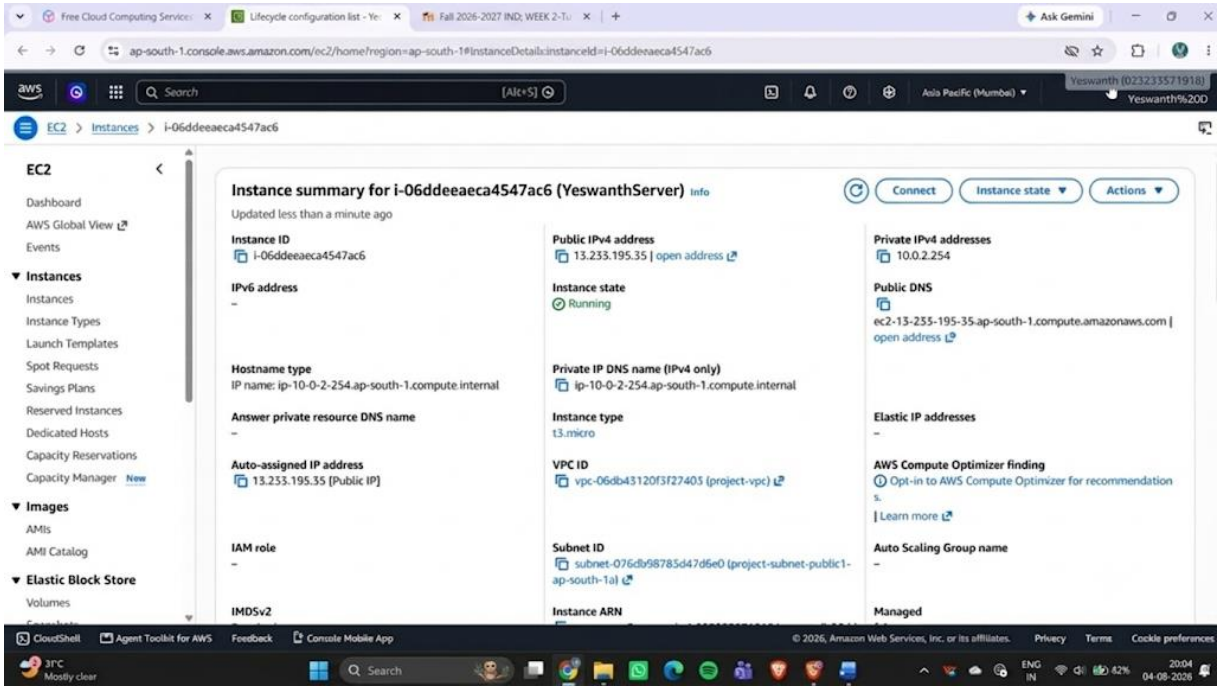
S3 Lifecycle Rule Configuration



EC2 Instance Launch



EC2 Instance Running



Difference between Amazon S3 Storage Classes (Standard vs Glacier)

Amazon S3 Standard is designed for frequently accessed data. It provides low latency, high availability (99.99%), and high durability (99.999999999%). It is suitable for active files such as websites, applications, and frequently used documents.

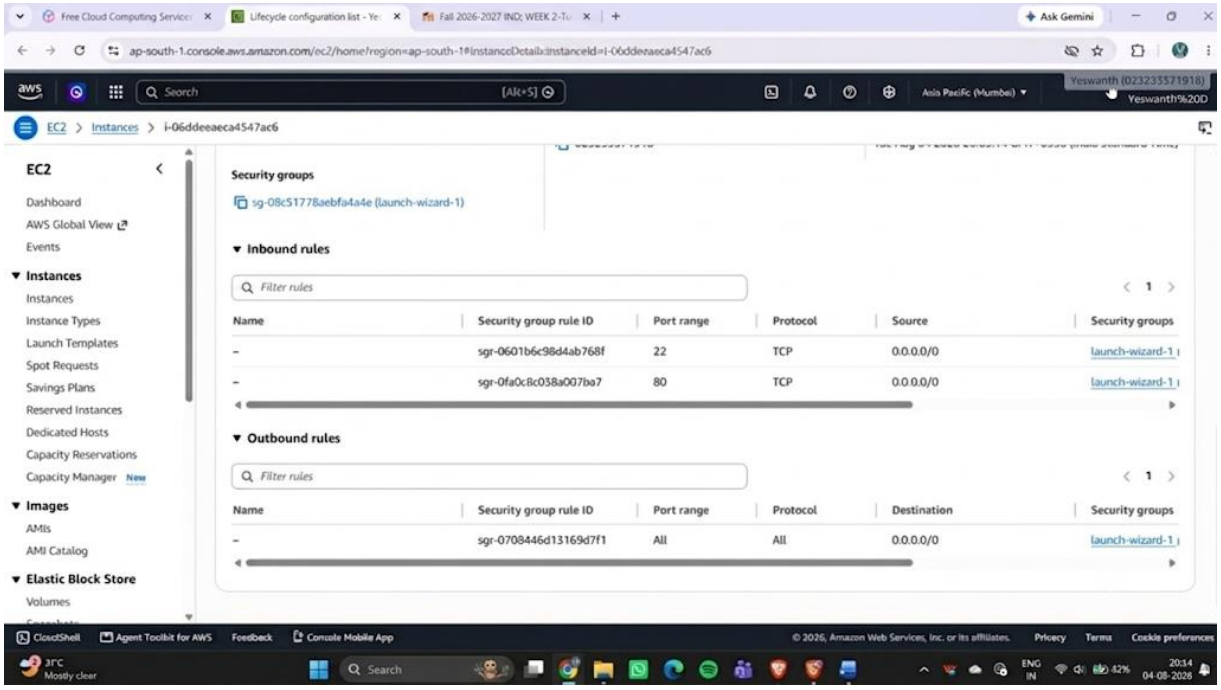
Amazon S3 Glacier is designed for long-term data archiving and backup. It offers a much lower storage cost than S3 Standard but has a higher retrieval time because archived data must be restored before it can be accessed. It is ideal for files that are rarely accessed but need to be retained for long periods.

S3 Standard	S3 Glacier
Frequently accessed data	Rarely accessed archival data
Immediate access	Retrieval may take minutes to hours
Higher storage cost	Lower storage cost
Suitable for active applications	Suitable for backup and archival

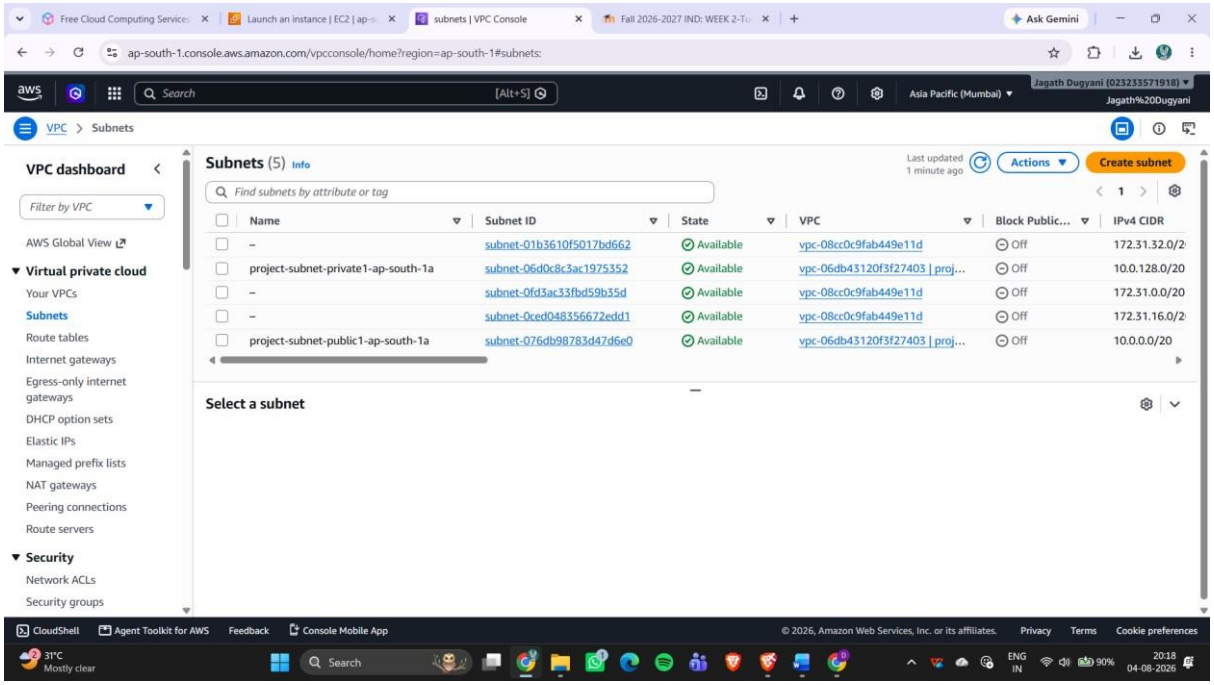
Why Lifecycle Policies are Useful for Cost Optimization

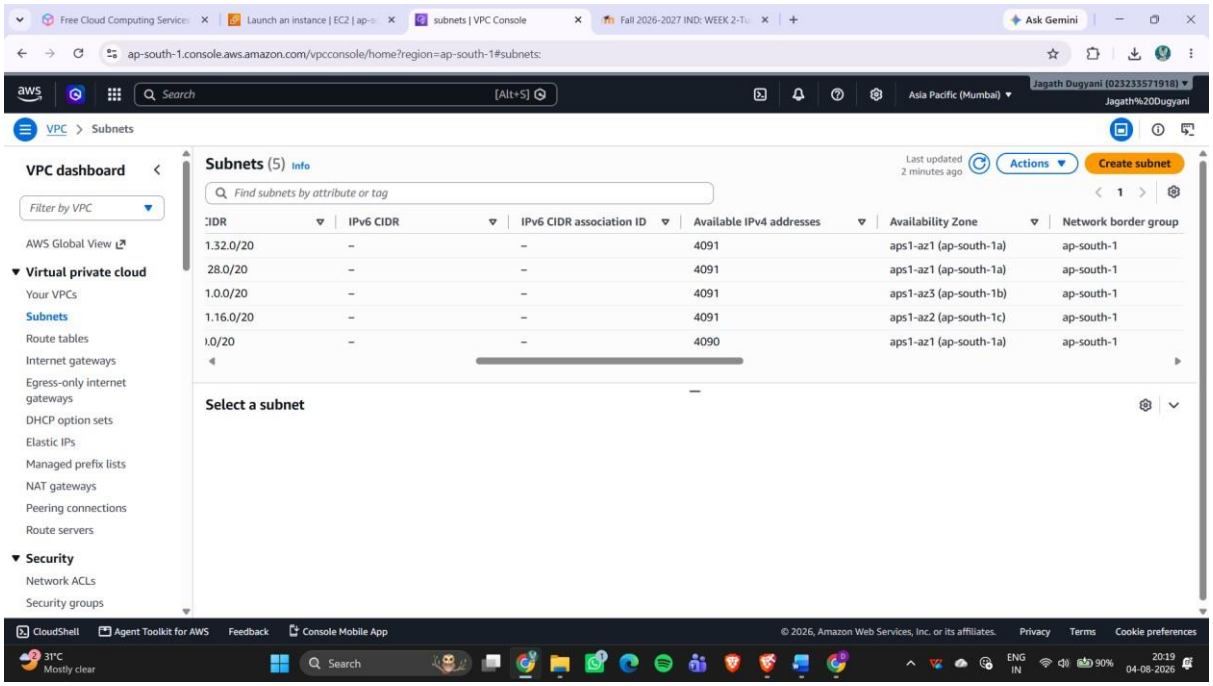
Amazon S3 Lifecycle Policies automatically move objects between storage classes or delete them after a specified period. This helps reduce storage costs by transferring infrequently accessed files from the Standard storage class to lower-cost storage classes such as Glacier. Lifecycle policies eliminate the need for manual management, improve storage efficiency, and ensure that data is stored in the most cost-effective manner throughout its lifecycle.

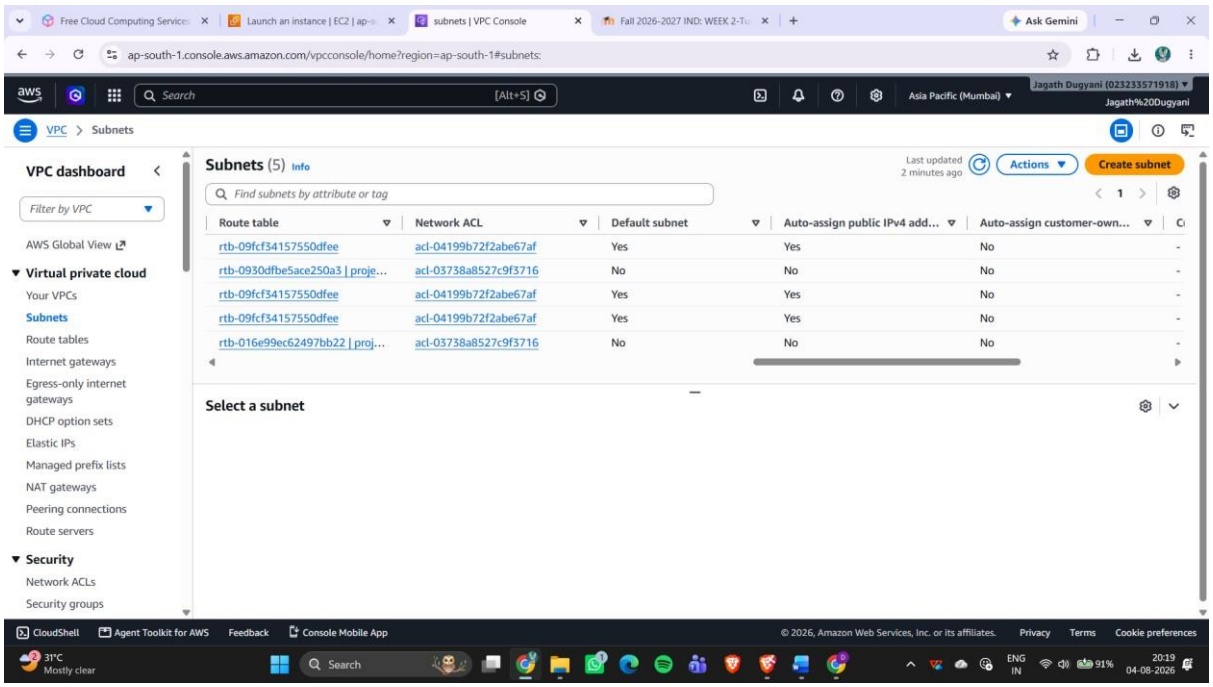
Security Group Rules

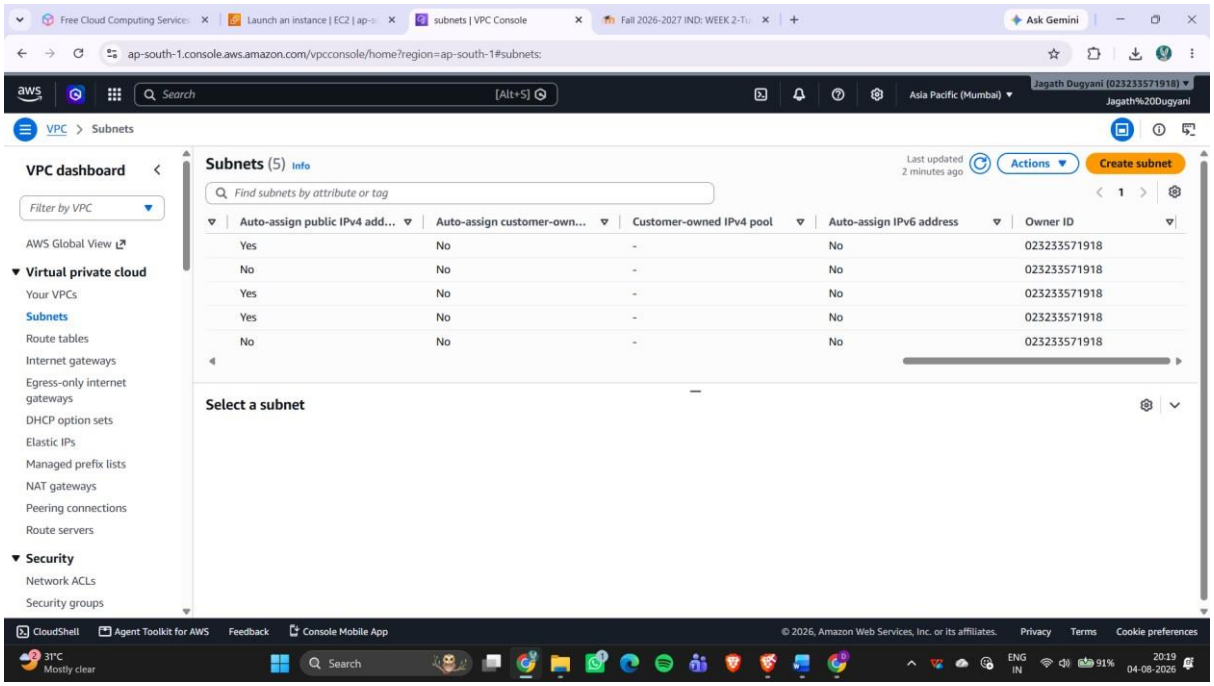


VPC subnets

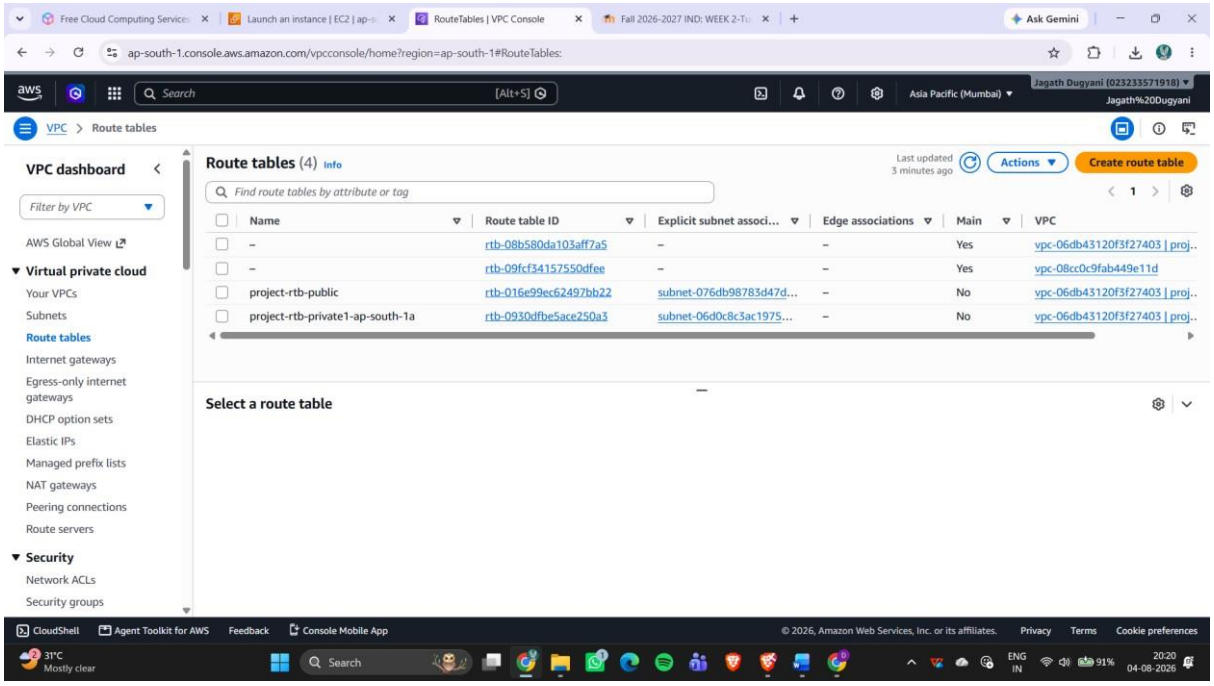


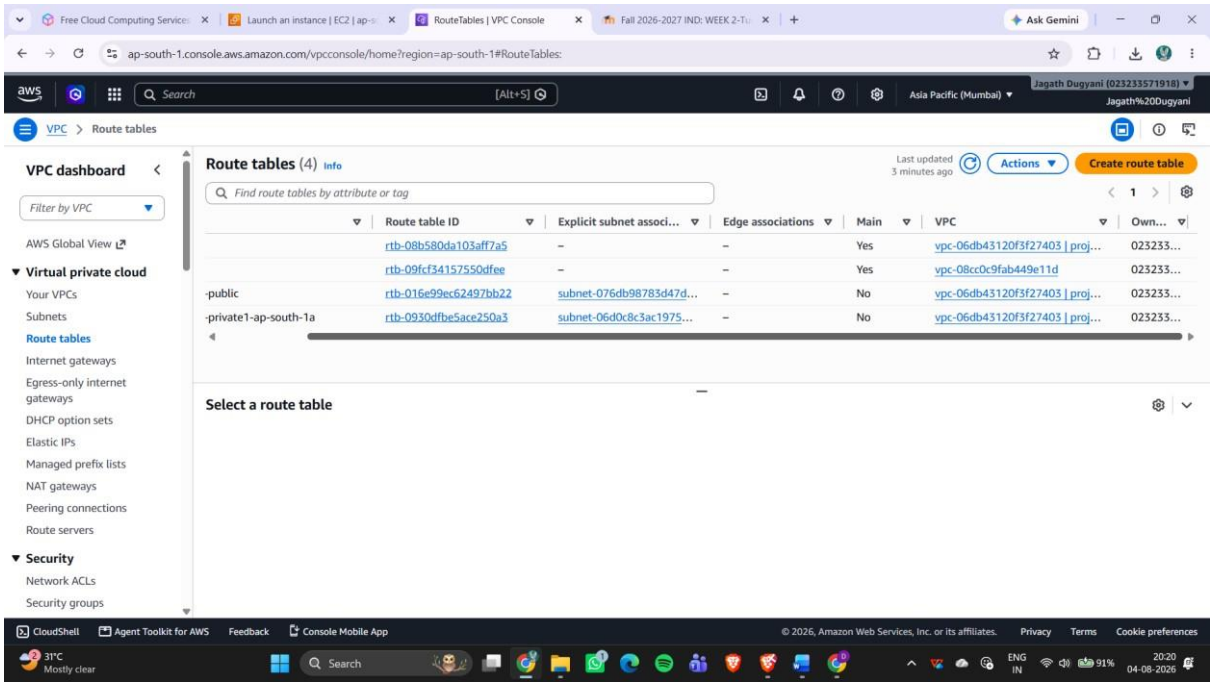

Free Cloud Computing Services
Launch an instance | EC2 | ap-s
subnets | VPC Console
Fall 2026-2027 IND: WEEK 2-Tu
Ask Gemini
ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#subnets:
Search
[Alt+S]
Asia Pacific (Mumbai)
Jagath Dugyani (023233571918)
Jagath%20Dugyani
VPC > Subnets
VPC dashboard
Filter by VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
Subnets (5) Info
Last updated 2 minutes ago
Actions
Create subnet
Find subnets by attribute or tag
1
:IDR
IPv6 CIDR
IPv6 CIDR association ID
Available IPv4 addresses
Availability Zone
Network border group
1.32.0/20 – – 4091 aps1-az1 (ap-south-1a) ap-south-1
28.0/20 – – 4091 aps1-az1 (ap-south-1a) ap-south-1
1.0.0/20 – – 4091 aps1-az3 (ap-south-1b) ap-south-1
1.16.0/20 – – 4091 aps1-az2 (ap-south-1c) ap-south-1
.0/20 – – 4090 aps1-az1 (ap-south-1a) ap-south-1
Select a subnet
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
31°C
Mostly clear
Search
ENG
IN
90%
20:19
04-08-2026


Free Cloud Computing Services
Launch an instance | EC2 | ap-s
subnets | VPC Console
Fall 2026-2027 IND: WEEK 2-Tu
Ask Gemini
ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#subnets:
Search
[Alt+S]
Asia Pacific (Mumbai)
Jagath Dugyani (023233571918)
Jagath%20Dugyani
VPC > Subnets
VPC dashboard
Filter by VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
Subnets (5) Info
Last updated 2 minutes ago
Actions
Create subnet
Find subnets by attribute or tag
1
Route table
Network ACL
Default subnet
Auto-assign public IPv4 add...
Auto-assign customer-own...
rtb-09fcf34157550dfee acl-04199b72f2abe67af Yes Yes No -
rtb-0930dfbe5ace250a3 | proje... acl-03738a8527c9f3716 No No No -
rtb-09fcf34157550dfee acl-04199b72f2abe67af Yes Yes No -
rtb-09fcf34157550dfee acl-04199b72f2abe67af Yes Yes No -
rtb-016e99ec62497bb22 | proj... acl-03738a8527c9f3716 No No No -
Select a subnet
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
31°C
Mostly clear
Search
ENG
IN
91%
20:19
04-08-2026

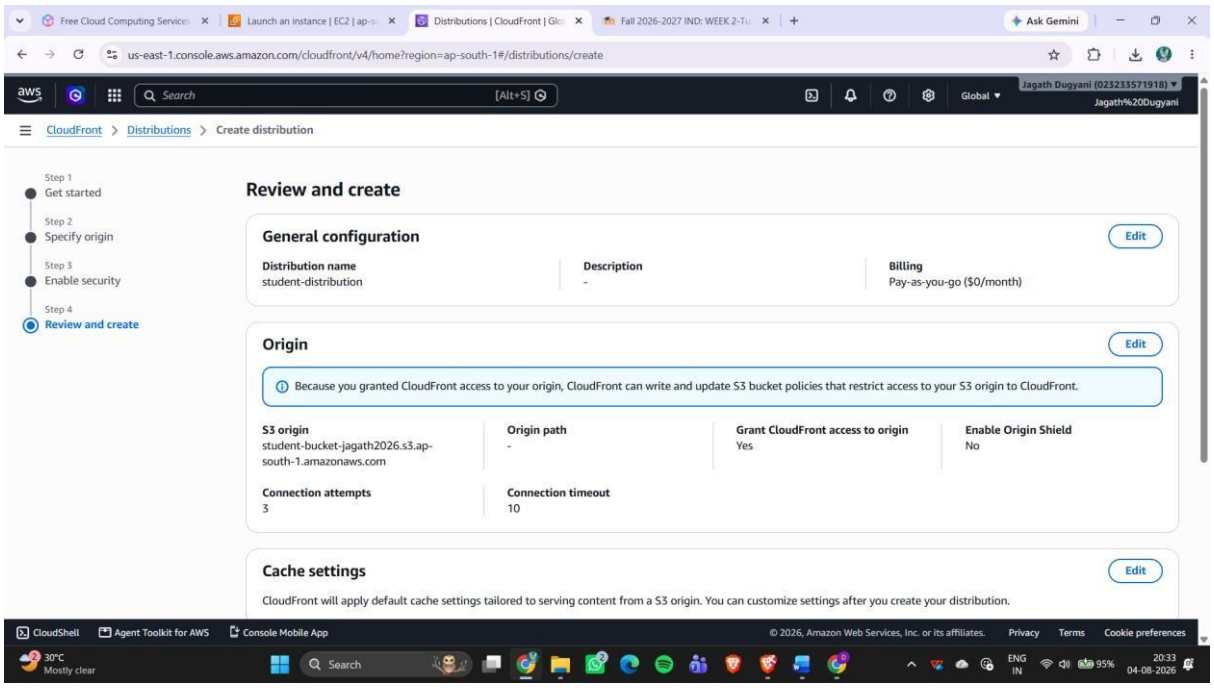


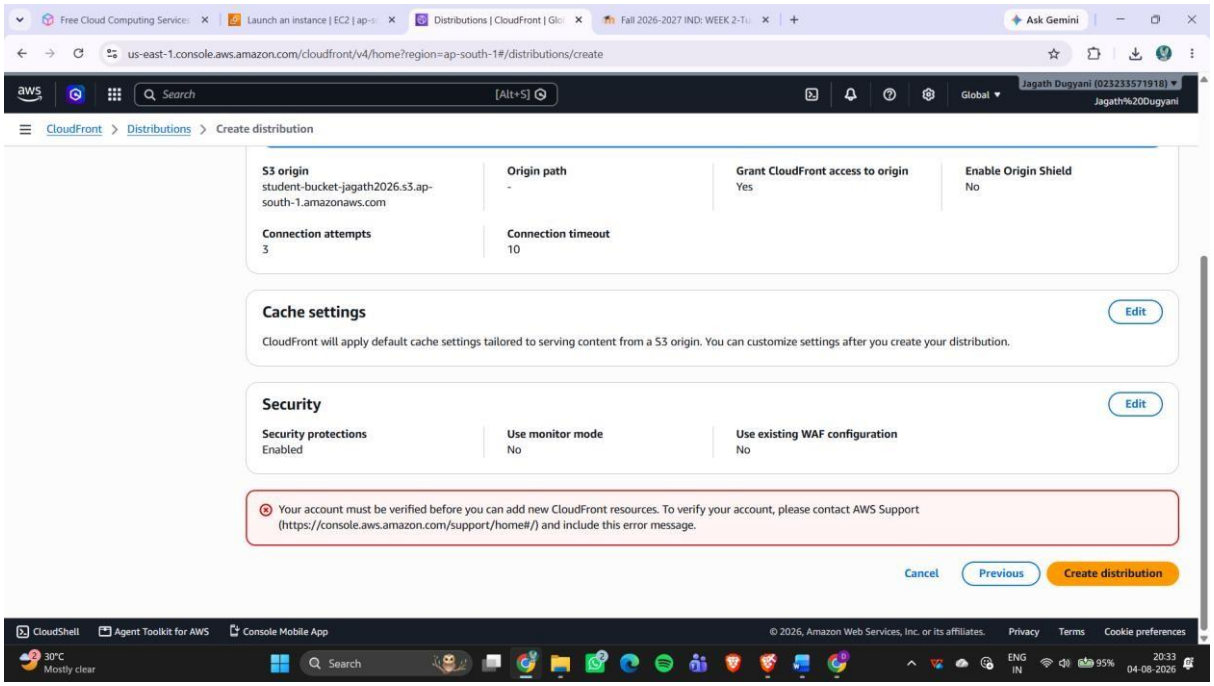
Route tables




ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#RouteTables:
VPC > Route tables
VPC dashboard
Route tables (4) Info
Actions
Create route table
Route table ID
Explicit subnet associ...
Edge associations
Main
VPC
rtb-08b580da103aff7a5
rtb-09fcf34157550dfee
rtb-016e99ec62497bb22
rtb-0930dfbe5ace250a3
Select a route table

Cloud front


us-east-1.console.aws.amazon.com/cloudfront/v4/home?region=ap-south-1#/distributions/create
CloudFront > Distributions > Create distribution
Review and create
General configuration
Edit
Distribution name
student-distribution
Description
Billing
Pay-as-you-go ($0/month)
Origin
Because you granted CloudFront access to your origin, CloudFront can write and update S3 bucket policies that restrict access to your S3 origin to CloudFront.
S3 origin
student-bucket-jagath2026.s3.ap-south-1.amazonaws.com
Origin path
Grant CloudFront access to origin
Yes
Enable Origin Shield
No
Connection attempts
3
Connection timeout
10
Cache settings
CloudFront will apply default cache settings tailored to serving content from a S3 origin. You can customize settings after you create your distribution.



Note: CloudFront distribution creation could not be completed because AWS requires account verification before new CloudFront resources can be created. The configuration steps were completed until the final creation stage, where AWS blocked the request due to account restrictions.

Role of Security Groups vs Route Tables

Security Groups act as virtual firewalls for AWS resources such as EC2 instances. They control which inbound and outbound traffic is allowed based on protocols, ports, and IP addresses. Security Groups are stateful, meaning return traffic is automatically allowed.

Route Tables determine where network traffic is directed within a VPC. They contain routing rules that specify the destination of packets, such as the Internet Gateway, NAT Gateway, or other subnets. Route Tables do not filter traffic; they only define the path that traffic follows.

How CloudFront Improves Content Delivery Performance

Amazon CloudFront is a Content Delivery Network (CDN) that caches content at edge locations around the world. When a user requests content, it is served from the nearest edge location instead of the original server, reducing latency and improving load times. CloudFront also provides high availability, automatic scaling, and enhanced security through HTTPS support and protection against DDoS attacks, resulting in faster and more reliable content delivery.